

Target bring-up runbook — Ryzen 5 9600X + ASUS B650E-PLUS WIFI + RX 7900 XT

From cables to running LuicipherOS. Assumes the install USB holds our netinstall ISO (see `iso/README.md`) and the repo is pushed.

0. Physical (before first power-on)

- ☐ Inventory vs Alza list; photo RAM + PSU labels.
- ☐ GPU: 2x SEPARATE PCIe 8-pin cables (no daisy-chain pigtail).
- ☐ Clearances: GPU $\leq$ 330 mm, cooler $\leq$ 160 mm, 2nd SSD mounted if present.
- ☐ Monitor -> GPU's own port (HDMI ok for install; DP cable later for high refresh). NOT the motherboard HDMI (weak iGPU).
- ☐ Keyboard, mouse, install USB inserted.

1. BIOS (Del on boot, F7 Advanced Mode)

Report first: BIOS version+date ($\geq$ mid-2024 AGESA?), CPU string, RAM capacity/speed as detected.

Path	Setting
Ai Tweaker → Ai Overclock Tuner	EXPO I (not Tweaked)
Advanced → PCI Subsystem Settings	Above 4G Decoding: Enabled, then Re-Size BAR: Enabled
Advanced → CPU Configuration	SVM Mode: Enabled (VM dev future)
Boot → CSM	Disabled (pure UEFI)
Boot → Secure Boot	Disabled / Other OS (systemd-boot unsigned)
Boot → Fast Boot	Disabled (keeps USB boot visible)
Boot Option #1	Install USB

Leave: fTPM default, PBO/curve (weeks later), fan curves (default). F10 Save & Exit.

2. Live boot + preflight

```
ls /sys/firmware/efi && echo EFI-OK
git clone https://github.com/luido/LuicipherOS.git ~/LuicipherOS && cd
~/LuicipherOS
git pull
cp iso/airootfs/usr/local/bin/luicipheros-install /usr/local/bin/luicipheros-
install
./install/preflight.sh --target    # must PASS; fix FAILs first
```

3. Disks (READ before writing — 2 SSDs likely)

```
lsblk -d -o NAME,SIZE,MODEL
```

- System disk is usually `/dev/nvme0n1`; with two SSDs, match by SIZE. Script defaults to `nvme0n1` — speak up if yours differs. Wrong disk = wiped data disk. No guessing.
- `btrfs` size (manual until issue #2): set JSON value to (disk GiB - 2).

4. Install

```
luicipheros-install # no --vm. creds prompt, keyring ensure automatic
```

~5-10 min on NVMe. Reboot into the disk (fix boot order), log in as luci.

5. Bootstrap (FULL: Flatpaks included on real hardware)

```
git clone https://github.com/lui-do/LuicipherOS.git && cd LuicipherOS
sudo -v; while true; do sudo -n true 2>/dev/null || break; sleep 45; done &
./install/bootstrap.sh
kill %1 2>/dev/null
```

Snapshot hint (ADR-0001): `snapper/timeshift` checkpoint before first sync.

6. Verify

Desktop five: bar renders, SUPER+T Tabby, SUPER+Space Vicinae, SUPER+N swaync, Print region -> Gradia editor. ROCm: `rocminfo`, `python3 -c "import torch; print(torch.cuda.is_available())"` expect True, Blender HIP device, ollama run llama3.2:3b "1+1". Office: `lpstat -p`, LibreOffice opens a .docx. Portals (Flatpak file upload): from a Flatpak app (e.g. Slack), attach a file — the GTK file dialog MUST open. If not: `XDG_CURRENT_DESKTOP` should be Hyprland, and `systemctl --user status xdg-desktop-portal*` must show the hyprland + gtk backends. (yazi can NOT serve upload dialogs — portal picker only; yazi stays the keyboard file manager.) Fonts: `yay -Ss ttf-ms-fonts ttf-aptos` -> add winners to lists.

Green across = close issue #1.

7. If stuck

- No EFI -> firmware row back to UEFI (VM lesson, same on hardware).
- `archinstall` JSON errors -> AGENTS.md field notes (sector_size, units, esp flag) + `python3 -m json.tool` validation.
- Mirror flakes -> rewrite `/etc/pacman.d/mirrorlist` with geo + `pacman -Syy`; keyring trouble -> `SigLevel=Never` rescue (see wiki 04).
- AUR build stalls -> `yay v13 --answer*` None flags are in `scripts/ensure-packages.sh`; source builds OOM small machines (prefer *-bin).